Cardiovascular System (CARD) · NPLEX-P1-CARD-06

CV-6 Cardiac Output and Its Regulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-06-01-C01				
NPLEX-P1-CARD-06-01-C02				
NPLEX-P1-CARD-06-01-C03				
NPLEX-P1-CARD-06-01-C04				
NPLEX-P1-CARD-06-01-C05				
NPLEX-P1-CARD-06-01-C06				
NPLEX-P1-CARD-06-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C01-BD1

Write out the output and pressure equations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C01 Activity: NPLEX-P1-CARD-06-01-C01-BD1 Date: _____

Without looking at your notes, write out the equations for cardiac output, stroke volume, ejection fraction, cardiac index, mean arterial pressure from cardiac output and resistance, mean arterial pressure from a cuff reading, and cardiac output by the Fick principle. Beside each equation write the normal resting value of every quantity in it, with units. Then write the full formula for the oxygen content of blood, label which part is carried by hemoglobin and which is dissolved, and write how many millilitres per decilitre the dissolved part contributes. Write why the cuff rule for mean arterial pressure uses one third of the pulse pressure and when that rule underestimates the true value. Finish by writing where arterial and mixed venous samples must be drawn and why mixed venous blood has to come from that site.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C01-BD2

Work one patient two ways and read the venous saturation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C01 Activity: NPLEX-P1-CARD-06-01-C01-BD2 Date: _____

A resting patient has a heart rate of 60 bpm, an end-diastolic volume of 150 mL and an end-systolic volume of 90 mL. Calculate stroke volume, cardiac output and ejection fraction, and state whether each one is normal. The same patient has an oxygen consumption of 240 mL/min, hemoglobin of 12 g/dL, arterial saturation of 97 percent with an arterial PO2 of 90 mmHg, and pulmonary artery saturation of 55 percent with a PO2 of 30 mmHg. Calculate cardiac output by the Fick principle, showing every step including the conversion from millilitres per decilitre to millilitres per litre. Then write what the low pulmonary artery saturation tells you about oxygen delivery relative to demand, and explain from the Fick equation why the arteriovenous oxygen difference widened. Finish by writing what a sample taken from a forearm vein instead of the pulmonary artery would do to your calculated output, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C02-BD1

Draw the Frank-Starling curve and define the three determinants

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C02 Activity: NPLEX-P1-CARD-06-01-C02-BD1 Date: _____

Draw a set of axes with end-diastolic volume on the horizontal axis and stroke volume on the vertical axis, and draw the normal Frank-Starling curve on it. Mark the part of the curve the normal heart works on and write what limits further stretch. Then draw a second curve for raised contractility and a third for reduced contractility, and label which direction each one shifted. Separately, define preload, afterload and contractility, and for each one write the strict definition, the measurement used in practice as an index of it, and what the ventricle is doing physically. Write why a stiff ventricle can show a high filling pressure at a normal end-diastolic volume. Finish by writing the sarcomere length that gives maximum force and the two cellular reasons stretch increases force, naming which of the two matters more.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C02-BD2

Follow three single changes through to stroke volume

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C02 Activity: NPLEX-P1-CARD-06-01-C02-BD2 Date: _____

For each of the three changes below, write which of the three determinants of stroke volume it acts on first, then what happens to end-diastolic volume, end-systolic volume, stroke volume and ejection fraction, then how the Frank-Starling curve moves, then how the pressure-volume loop changes shape. Write each chain as a sequence of steps rather than single words. The three changes are a sudden rise in aortic pressure with venous return unchanged, a rapid infusion of one litre of saline, and an acute coronary occlusion that leaves part of the ventricle weakly contracting. For the aortic pressure change, explain over several beats why stroke volume partly recovers and what that recovery costs. Then write why ejection fraction is not a clean measure of contractility, and why a heart rate of 200 bpm lowers cardiac output rather than raising it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C03-BD1

Draw the cardiac function and venous return curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-BD1 Date: _____

Draw one set of axes with right atrial pressure on the horizontal axis and flow in litres per minute on the vertical axis. On it draw the cardiac function curve and the venous return curve, and mark the point where they cross. Write what that crossing point represents and why the two flows must be equal. Label the right atrial pressure at which the venous return curve meets the horizontal axis, and write what that pressure is called and its normal value. Write the equation for venous return, name what sets the slope of the curve, and write why the curve flattens at low right atrial pressures. Finish by writing what stressed volume means, roughly what fraction of the blood the veins hold, and what venoconstriction does to mean systemic filling pressure and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C03-BD2

Name the problem from cardiac output and right atrial pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-BD2 Date: _____

Four patients each have a low cardiac output. Patient A has a right atrial pressure of 1 mmHg, patient B has 14 mmHg, patient C has 2 mmHg after three days of vomiting, and patient D has 15 mmHg two days after a large anterior infarction. For each patient write whether the primary problem lies in the heart or in the vascular compartment, give the reasoning from whether the two numbers moved in the same direction or in opposite directions, name which curve shifted and in which direction, and give a cause from the notes that fits. Then write what happens on the diagram when a dilated exercising muscle bed or an arteriovenous fistula lowers total peripheral resistance, and say why cardiac output changes in that case while right atrial pressure stays about the same.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-06-01-C03-RT01

Rebuild the hemorrhage sequence on the curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from blood loss to a partly recovered cardiac output, one step per line and in order. Beside the step where mean systemic filling pressure falls, write why losing blood lowers it and which part of the blood volume is responsible. Beside the reflex steps, write which receptors sense the change and which effector action restores each of the two curves. Then draw one set of axes and show three things on it: the starting operating point, the point immediately after the blood loss, and the point after the reflex response. Beside the final point write why right atrial pressure stays low even though cardiac output has improved.

Rebuild from memory: Hemorrhage on the cardiac function and venous return diagram

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C04-BD1

Write the law of Laplace and the determinants of oxygen demand

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C04 Activity: NPLEX-P1-CARD-06-01-C04-BD1 Date: _____

Write the law of Laplace in the form used for the ventricle, name each symbol, and write what the quantity on the left side means physically. Write which phase of the cardiac cycle makes that quantity the afterload. Then write, in one line each, what happens to wall stress when pressure rises, when the chamber radius grows, and when the wall thickens, and explain the thickness effect in terms of how the load is shared. Separately, list the three main determinants of myocardial oxygen demand, write which kind of work costs the most oxygen, and write the simple bedside index of demand and the two numbers it multiplies. Finish by writing the percentage of delivered oxygen the resting heart already extracts, what that leaves as the only way to meet higher demand, and which part of the cardiac cycle carries most left ventricular coronary flow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C04-BD2

Remodeling and the ischemic cost of a faster rate

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C04 Activity: NPLEX-P1-CARD-06-01-C04-BD2 Date: _____

A patient's systolic pressure has run at 200 mmHg for years and the ventricular wall has thickened from 1.0 cm to about 1.7 cm with a normal cavity size. Write which overload this is, which measurement rose to signal it, how the myocytes added sarcomeres, and what the thickening did to wall stress. Then write the two costs of that adaptation, including what happens to filling pressure and which extra heart sound appears, and explain why the subendocardium becomes vulnerable to ischemia even when the coronary arteries are normal. Next, a second patient has severe mitral regurgitation and a large dilated ventricle. Write which overload this is, which wall stress rose, how the sarcomeres were added, and what the thickness to radius ratio does while compensation holds and when it fails. Finish with a third patient who has a fixed coronary stenosis and a heart rate that rises from 70 to 140 bpm: write the two separate reasons the faster rate causes ischemia, one on the demand side and one on the supply side.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-06-01-C04-CP01

Concentric against eccentric remodeling

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Concentric hypertrophy	Eccentric hypertrophy and dilation
Typical cause		
Which wall stress signals it		
How sarcomeres are added and what the cell shape does		
Thickness to radius ratio		
What happens to wall stress		
Effect on chamber stiffness and filling pressure		
Typical consequence and the extra heart sound		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C05-BD1

Draw the baroreceptor reflex arc

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-BD1 Date: _____

Draw the baroreceptor reflex as a labelled pathway from sensor to effector. Name the two receptor sites and the layer of the vessel wall the endings sit in, name the nerve each afferent travels in with its cranial nerve number, and name the medullary nucleus they all reach. Then draw the central connections: write which output rises and which falls when pressure rises, and name the two medullary regions that carry the sympathetic side. List every effector the reflex acts on and write what the reflex changes at each one. Finish by writing the pressure range over which carotid receptors fire, how the aortic receptors differ, what the receptors do within one to two days, and what that resetting means for long-term pressure control.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C05-BD2

Run the reflex forwards and backwards in four situations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-BD2 Date: _____

For each situation below write what happens to baroreceptor firing, to vagal output, to sympathetic output, and then to heart rate, contractility, arteriolar tone and venous tone, and finally to mean arterial pressure. The four situations are a patient who has lost about 10 percent of blood volume, a patient given a drug that constricts arterioles through alpha-1 receptors, a patient whose carotid sinus is massaged during a fast rhythm, and an older man who faints when a tight collar presses on his neck. For the blood loss patient also write why the skin is cool and pale, why urine output falls, why the pulse pressure narrows, and what arterial pressure itself is doing at that point. For the carotid massage patient write what happens at the atrioventricular node and why that can stop the rhythm. Finish by writing what is added below a mean arterial pressure of about 80 mmHg and what happens below about 60 mmHg.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-06-01-C05-RT01

Rebuild the reflex response to a fall in pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-RT01 Date: _____

Rebuild the six-step cycle that runs from a fall in arterial pressure back to pressure returning toward its set point, one step per line and in order, and draw it as a loop with arrows so the feedback closes. Beside the afferent step write the two cranial nerves and the nucleus they reach. Beside the central step write what happens to vagal output and to sympathetic output. Beside the effector step list the four things that rise. Then write beside the loop what makes this negative feedback rather than positive feedback. Underneath, write the same six steps again for a rise in arterial pressure, with every direction reversed.

Rebuild from memory: Baroreceptor reflex response to a fall in arterial pressure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C06-BD1

Build the volume hormone table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C06 Activity: NPLEX-P1-CARD-06-01-C06-BD1 Date: _____

On blank paper build a table with one row each for angiotensin II, aldosterone, antidiuretic hormone, atrial natriuretic peptide and B-type natriuretic peptide. Use the columns source, main stimulus for release, receptor and second messenger, and main actions, and fill every cell. Then write out the renin-angiotensin-aldosterone sequence as seven numbered steps from the stimulus to the rise in volume and pressure, naming the three stimuli for renin release, the cell that releases it, which step is rate limiting, where the converting enzyme is most abundant, and the other peptide that enzyme breaks down. Finish by writing which arteriole angiotensin II constricts more and what that preserves, and how long each of the three control systems takes to act: the reflex, angiotensin II, and aldosterone.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C06-BD2

Three patients, three hormone patterns

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C06 Activity: NPLEX-P1-CARD-06-01-C06-BD2 Date: _____

Patient A has lost a litre of blood. Patient B has a high plasma osmolality from poor water intake with a normal blood volume. Patient C has an aldosterone-secreting adrenal adenoma. For each patient write what happens to renin, angiotensin II, aldosterone, antidiuretic hormone and the natriuretic peptides, and give the stimulus that moved each one. Then write which patient retains mainly sodium with water and which retains mainly water, and explain that difference from what each hormone acts on in the nephron. For patient C, write why the blood pressure is high and the plasma potassium is low but the patient has no edema, and name the process that halts the volume expansion. Finish by writing why high natriuretic peptide levels in heart failure fail to clear the retained fluid, and why inappropriate antidiuretic hormone secretion causes a low plasma sodium rather than hypertension.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-06-01-C06-PR01

Predict the response to renal artery narrowing

A patient has a narrowed renal artery, so perfusion pressure reaching the juxtaglomerular cells of that kidney is low while systemic arterial pressure is normal. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to renin release from the juxtaglomerular cells of the affected kidney, and name the two stimuli at that cell other than sympathetic input		
What happens to circulating angiotensin II		
What happens to the tone of the efferent arteriole compared with the afferent arteriole, and what that preserves		
What happens to aldosterone secretion and to the number of epithelial sodium channels in the principal cells		
What happens to urinary potassium and hydrogen ion secretion as sodium reabsorption rises		
What happens to plasma volume and to systemic arterial pressure over the next days		
What happens to renin release once systemic pressure has risen, and why the affected kidney keeps releasing it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C07-BD1

What standing does and how the body answers it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C07 Activity: NPLEX-P1-CARD-06-01-C07-BD1 Date: _____

Write the numerical definition of orthostatic hypotension, including both pressure criteria, the time limit, and the tilt angle that counts. Then write what standing does to the circulation as a sequence: the venous pressure at the foot in a motionless adult, how much blood pools and where, what filters into the leg tissue over minutes, and what happens in turn to venous return, stroke volume, cardiac output and mean arterial pressure. Next write the four things that normally defend pressure on standing, naming the reflex, the expected heart rate rise in beats per minute, the vessels that constrict, the mechanical pump and the structures that make it work in one direction, and the two hormones that add support over minutes. Finish by writing the mean arterial pressure at head level below which symptoms appear, listing the symptoms, and naming the symptom pattern specific to the neurogenic forms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C07-BD2

Separate the mechanisms from two sets of vital signs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C07 Activity: NPLEX-P1-CARD-06-01-C07-BD2 Date: _____

Patient A goes from 112/70 mmHg with a heart rate of 92 bpm lying down to 88/62 mmHg with a heart rate of 128 bpm at three minutes standing. Patient B goes from 162/88 mmHg with a heart rate of 96 bpm to 118/70 mmHg with a heart rate of 100 bpm. For each patient calculate the systolic fall and the heart rate rise, state whether the definition is met, write which limb of the reflex arc is working and which is not, and name the mechanism. Write which single number separates the two mechanisms and why that number behaves the way it does in each. Then write why patient B is hypertensive lying down, and in a patient with long-standing diabetes write the three nerve injury mechanisms from the notes and why a fixed fast resting heart rate can appear before the orthostatic drop. Finish by writing how postural tachycardia syndrome and vasovagal syncope each differ from both of these patients, using the heart rate and the pressure in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-06-01-C07-CA01

Lightheadedness on standing after two weeks in bed

A 74-year-old woman is seen on the ward on the fifteenth day of an admission for a hip fracture, during which she has been in bed almost continuously. She reports that each time she is helped to her feet she feels lightheaded, her vision dims, and her neck and shoulders ache, and twice she has had to be lowered back onto the bed. She takes no medication that blocks adrenergic receptors, has no diabetes, and has had no vomiting, diarrhoea or bleeding. Temperature is 36.8 °C (98.2 °F) and respirations are 16/min. Lying down, blood pressure is 128/76 mm Hg and pulse is 78/min. At three minutes standing, blood pressure is 96/60 mm Hg and pulse is 104/min. Her weight is 3 kg lower than on admission, her plasma sodium is 139 mEq/L (reference range 135 to 145 mEq/L), and her ratio of blood urea nitrogen to creatinine is 14 (reference range 10 to 20).

1. Recognize. State whether the pressure readings meet the definition of orthostatic hypotension, showing the systolic and diastolic falls and the heart rate rise you calculated.

2. Explain. Explain the mechanism by which two weeks in bed lowers plasma volume, naming the receptors stretched, the two hormones that change and in which direction, and write what happens to the size and stiffness of the ventricle and to the strength of the leg muscle pump.

3. Differentiate. Write how this presentation differs from orthostatic hypotension caused by volume loss through vomiting or bleeding and from orthostatic hypotension caused by failure of the sympathetic efferent pathway, naming the feature in these readings that separates it from the second of those.

4. Investigate. Write which measurements and bedside findings you would use to confirm the mechanism and what value or appearance each one would have, including what the heart rate response, the supine pressure and the urea to creatinine ratio should look like.

5. Decide. State which of the four normal defences against standing have been weakened in this patient, and which single one of them remains intact.

6. Justify. Justify your answer from the venous return side of the circulation: explain the fall in mean systemic filling pressure, what that does to the venous return curve and to the operating point, and why the intact reflex produces a brisk heart rate rise yet cannot hold the pressure up.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-06-01-C01-Q01

Reading the arteriovenous oxygen difference

A 62-year-old man is evaluated for shortness of breath. His oxygen consumption is 250 mL/min and is unchanged from a study done one year earlier. Arterial oxygen content is 20 mL/dL on both studies. Mixed venous oxygen content, sampled from the pulmonary artery, has fallen from 15 mL/dL last year to 12 mL/dL today. Hemoglobin concentration and arterial saturation are unchanged. Which of the following best describes this patient's cardiac output today compared with last year?

A. It has risen from about 5.0 L/min to about 6.3 L/min, because tissues are extracting less oxygen

B. It is unchanged at about 5.0 L/min, because oxygen consumption and arterial content are unchanged

C. It has fallen from about 5.0 L/min to about 3.1 L/min, because each litre of blood now gives up more oxygen

D. It cannot be estimated, because mixed venous content does not appear in the Fick relationship

Why did you pick it?

Board-style question NPLEX-P1-CARD-06-01-C04-Q02

Wall stress in a dilating ventricle

A 58-year-old woman with a dilated cardiomyopathy has a left ventricular end-diastolic radius that has increased by 40 percent over two years, while wall thickness and systolic pressure have not changed. Which of the following best explains why the oxygen cost of each beat has risen?

A. Systolic wall stress has risen in proportion to the radius, because wall stress varies with pressure times radius divided by twice the thickness

B. Oxygen extraction by the myocardium has fallen below its resting value of about 25 percent

C. The larger chamber lowers systolic wall stress but raises the volume of blood that must be ejected, and volume work costs more oxygen than pressure work

D. Coronary flow during systole has increased to match the larger chamber, raising the oxygen cost of the beat itself

Why did you pick it?